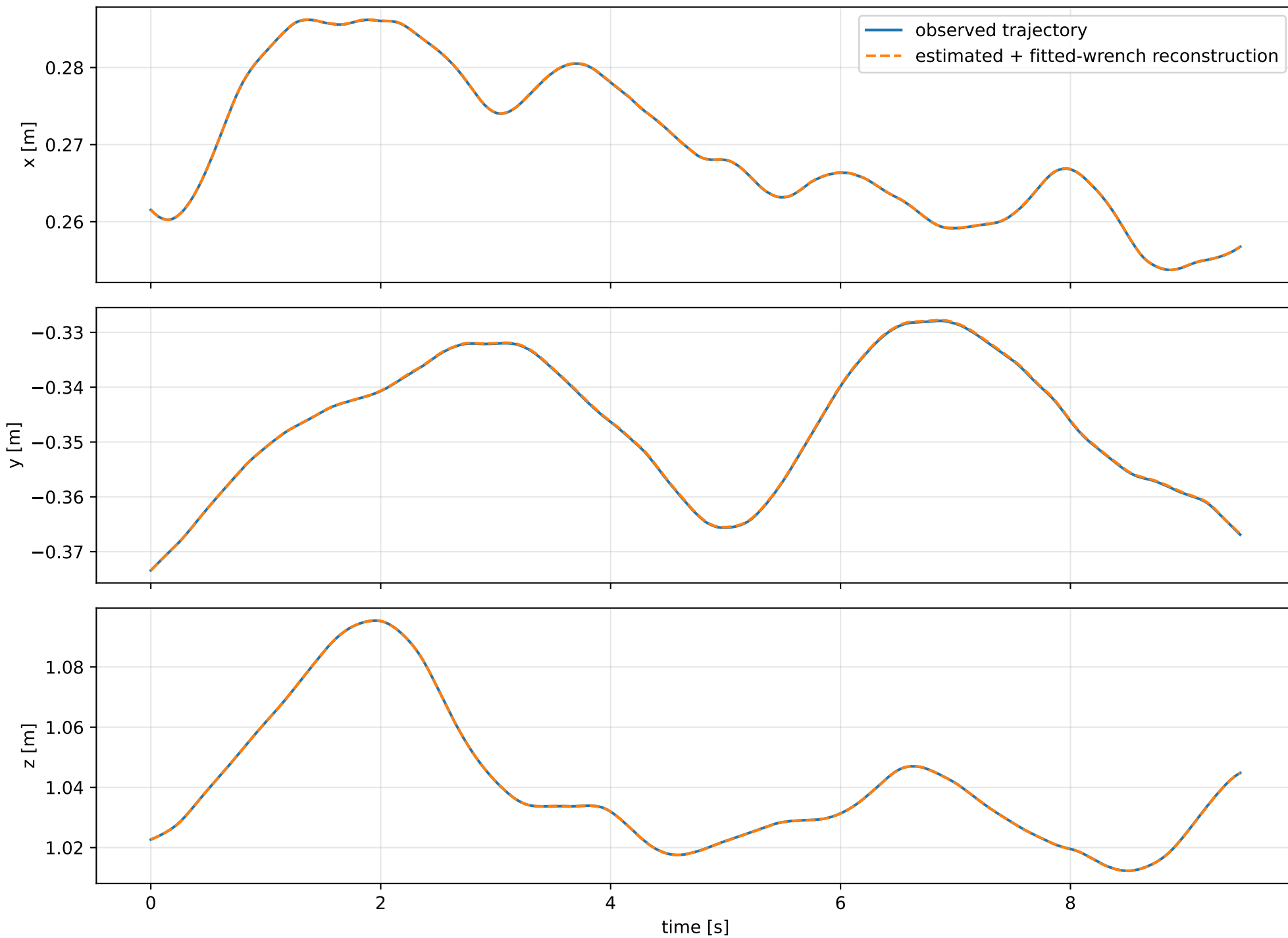
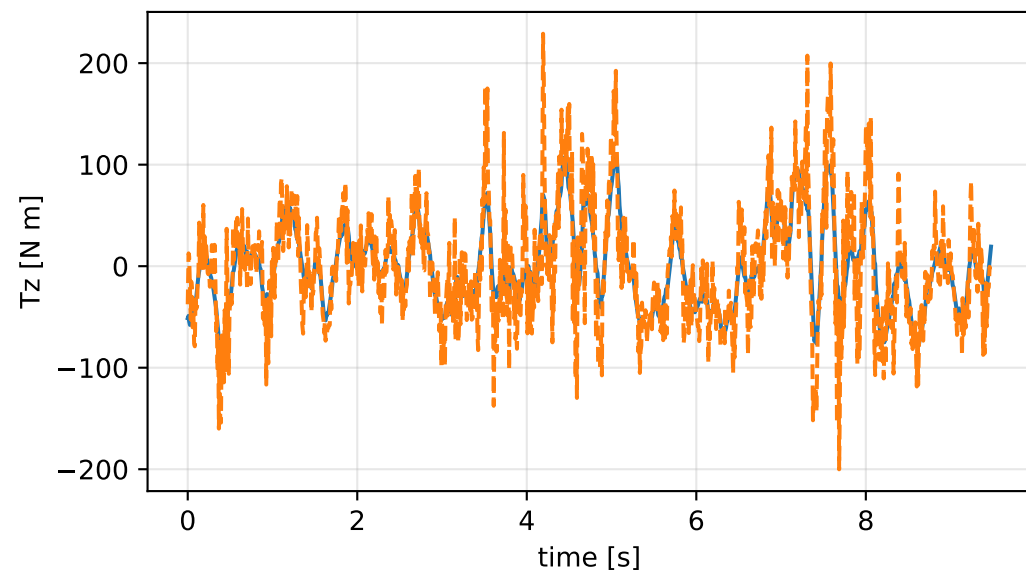
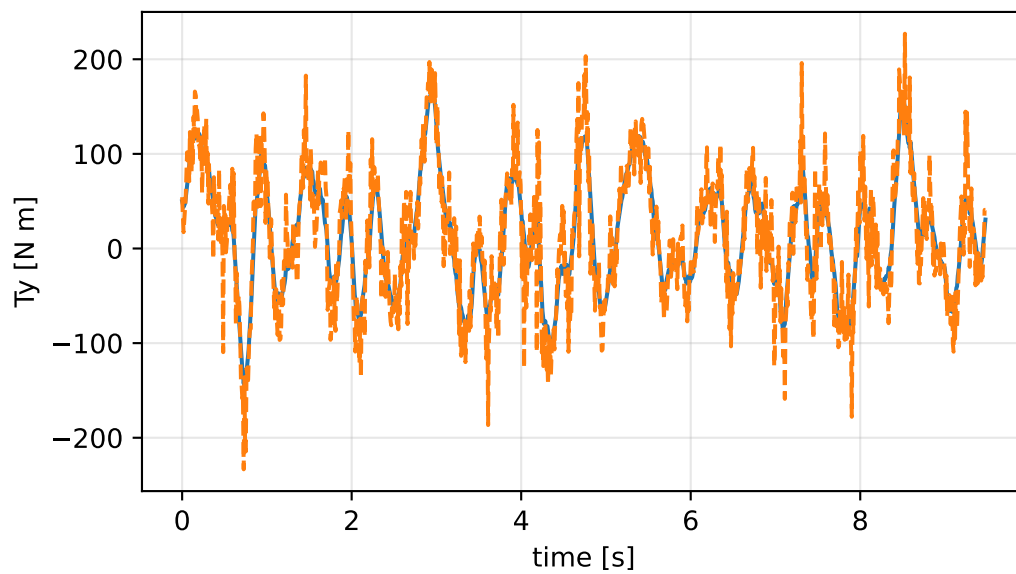
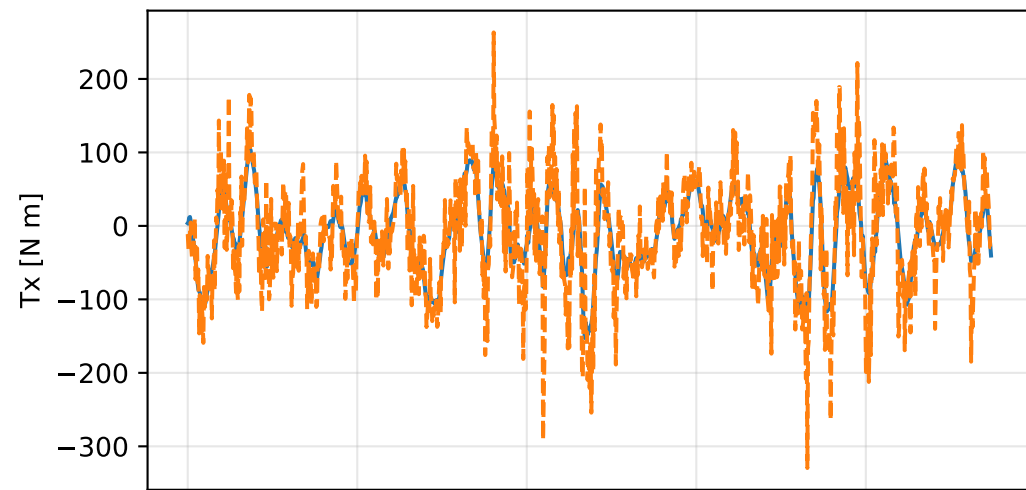
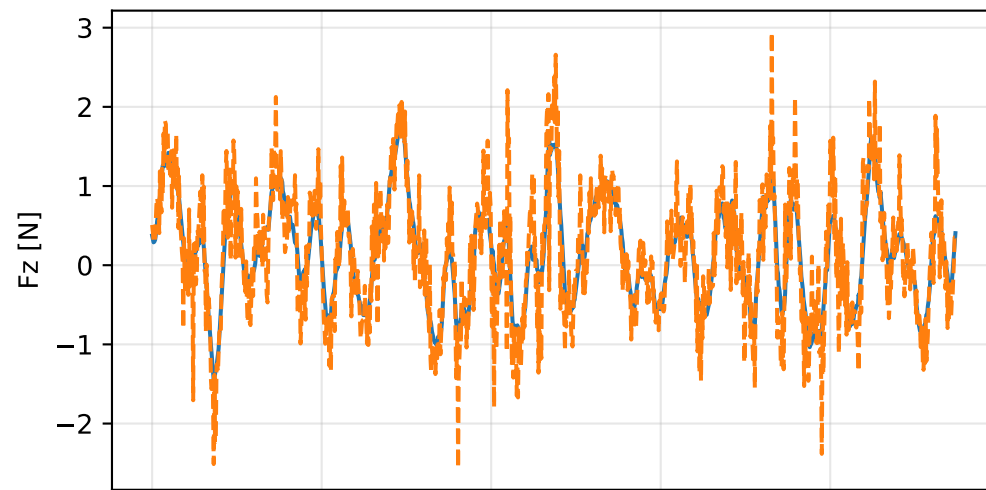
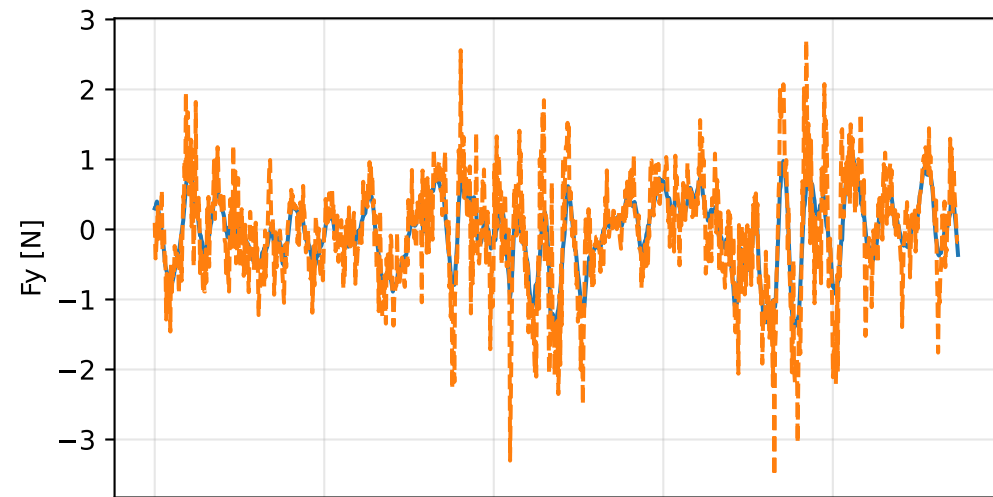
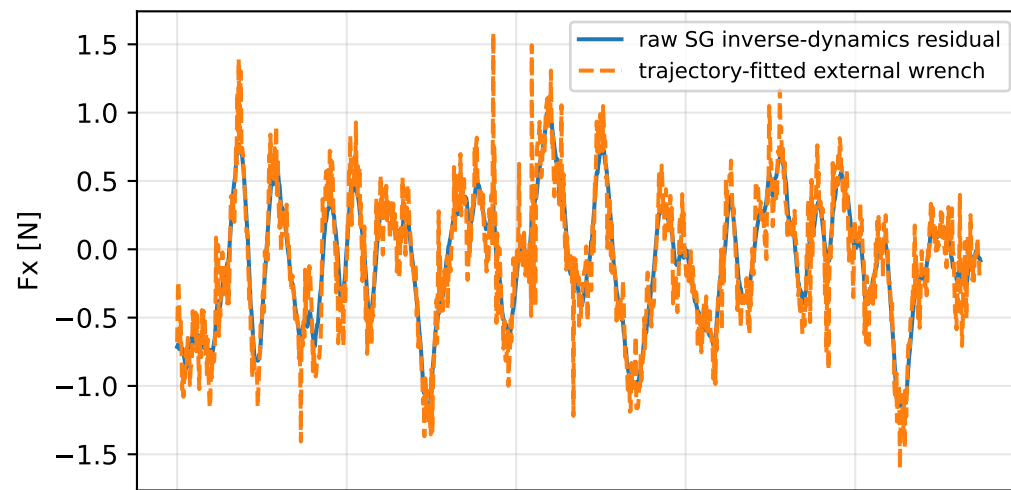


jacobian_finite_difference: trajectory

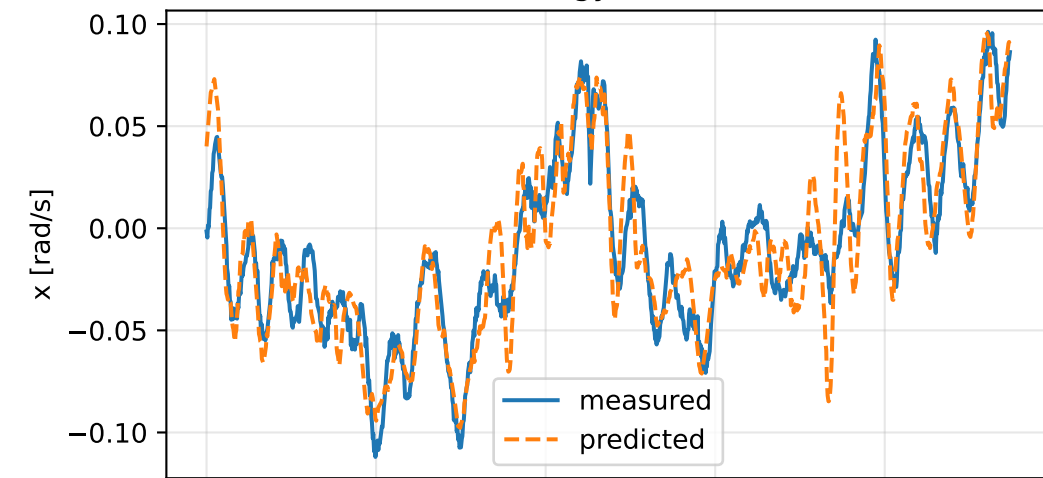


jacobian_finite_difference: residual wrench

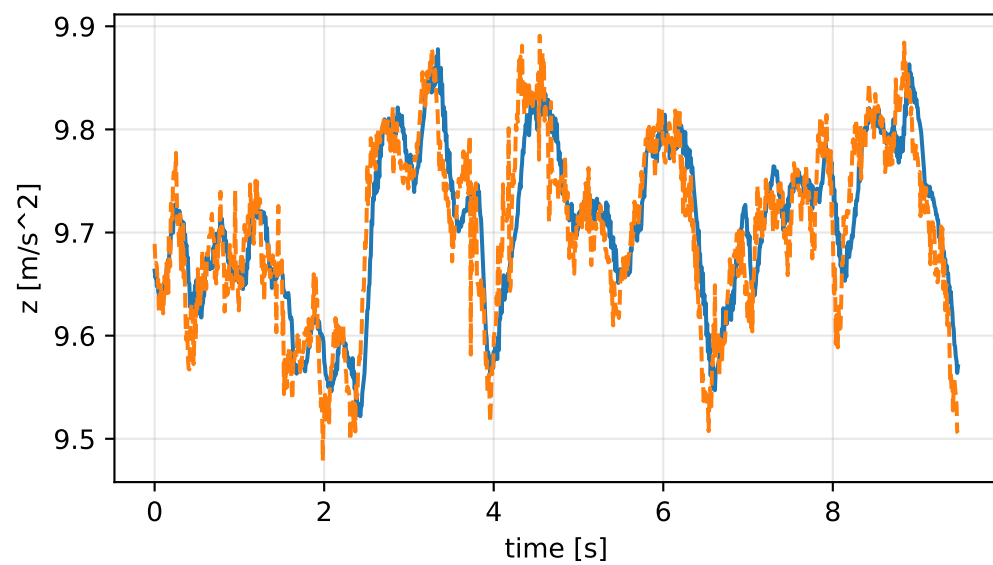
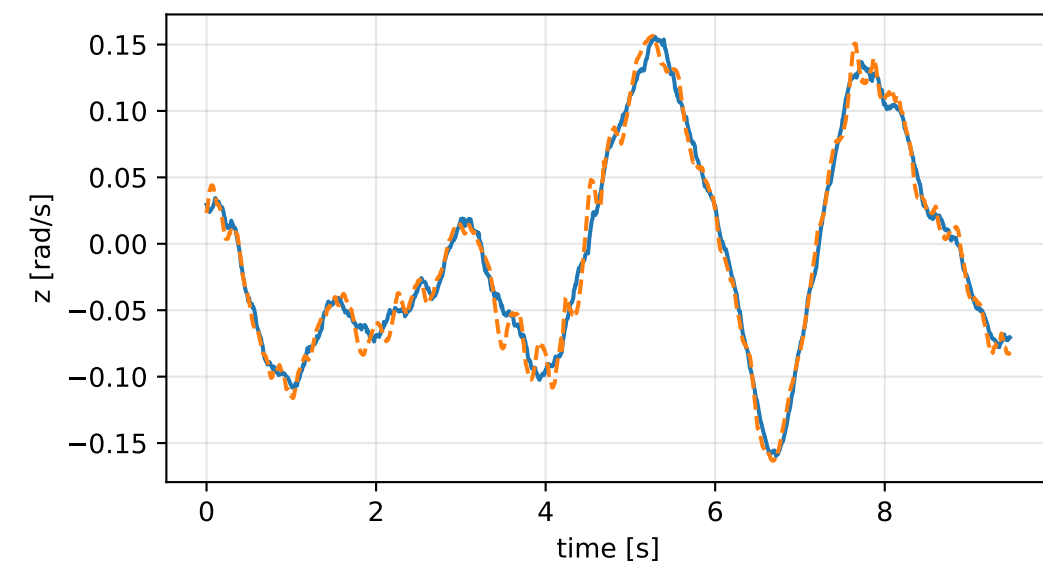
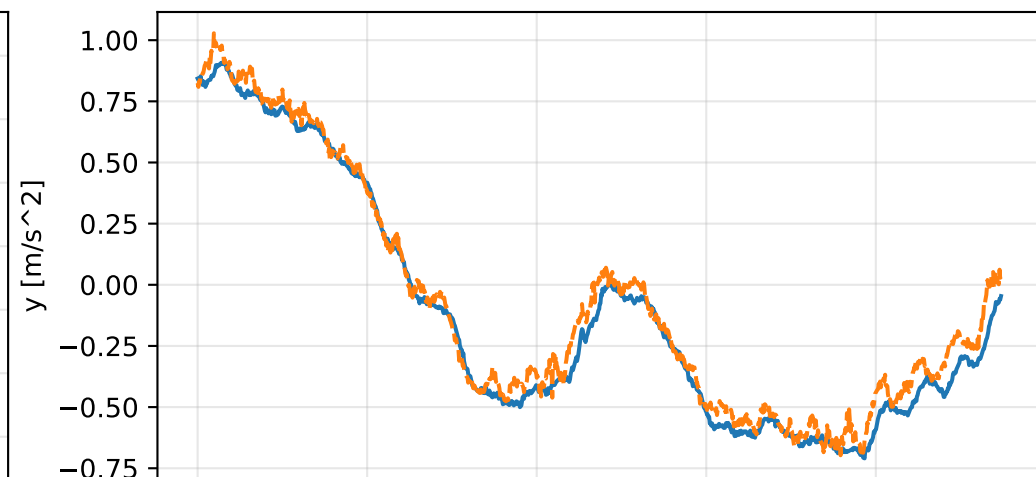
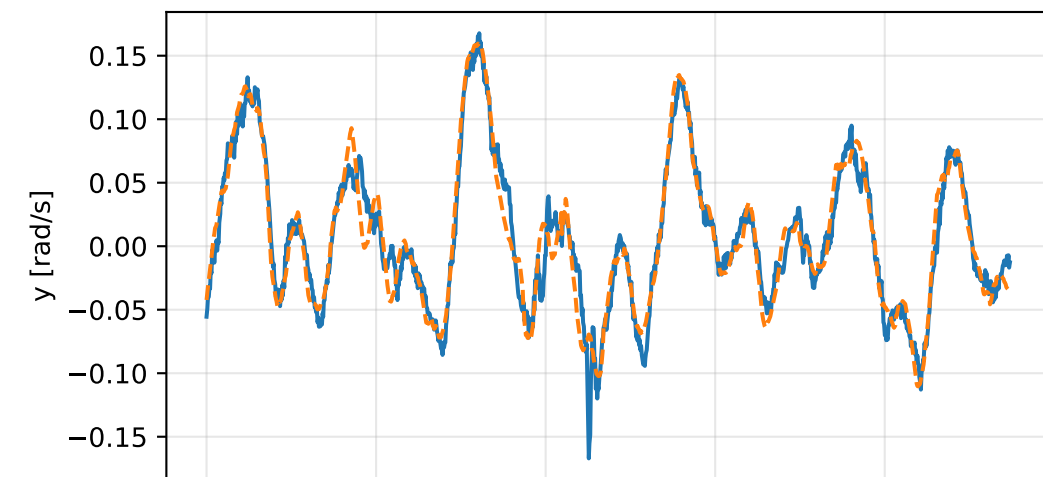
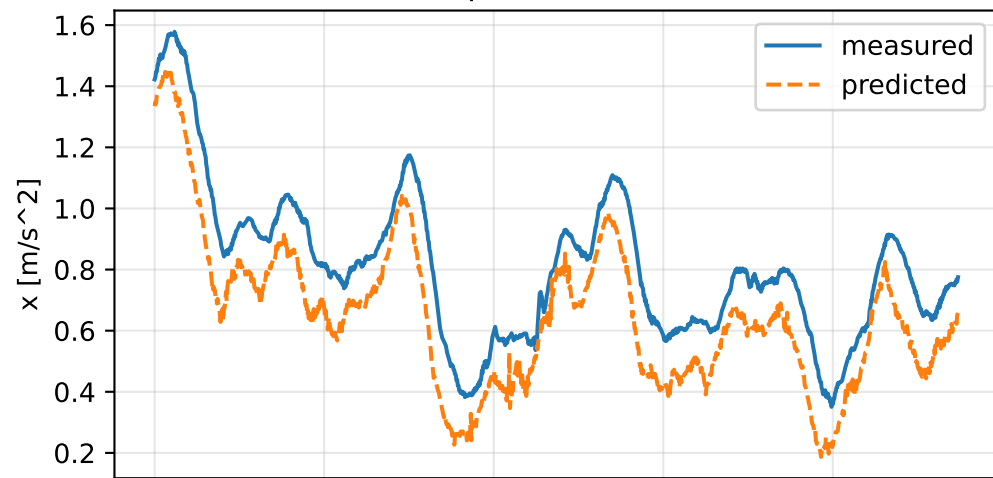


jacobian_finite_difference: measured sensor comparison

gyro



specific force



jacobian_finite_difference: nominal -> estimated parameters

Case: jacobian_finite_difference

status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 1.07794502 delta -1.27365257

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [117.1064457 -69.5375616 -73.7807669] d=[117.0414456 -69.5375609 -73.7807669]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [-69.5375616 149.8364351 -58.3746705] d=[-69.5375609 149.7714824 -58.3746705]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-73.7807669 -58.3746705 142.9383399] d=[-73.7807859 -58.3746706 142.8093399]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-1.3324326 -0.9585076 -1.1045125] d=[-1.3304079 -0.9584771 9.5097496e-03]

rotor force effectiveness

[1. 1. 1. 1.] -> [0.2327574 0.2460651 0.4124687 0.5627074] d=[-0.7672426 -0.7539349 -0.5875313 -0.4372926]

rotor lag [s] 0.199999857

gimbal lag [s] 0.0503390711